Chinese Remainder Theorem (CRT)

Beginner-friendly notes • Modular arithmetic • Worked examples • Python • Cryptography CTFs

Learning goal: Understand the notation, the theorem, when it applies, how to solve problems by hand, and how to verify a solution in Python.

1. Start with division and remainders

When an integer is divided by another integer, we get a quotient and a remainder. For example, 17 divided by 5 gives quotient 3 and remainder 2, because 17 = 5 × 3 + 2.

```
17 = 5 × 3 + 2 ⇒ 17 mod 5 = 2
```

a mod n means the remainder when a is divided by n. In Python, the remainder operator is written %. Examples: 14 mod 5 = 4; 20 mod 5 = 0; 23 mod 5 = 3.

2. What does the congruence symbol mean?

The notation **x ≡ a (mod n)** is read “x is congruent to a modulo n.” It means x and a leave the same remainder when divided by n. Equivalently, n divides x – a.

```
x ≡ a (mod n) ⇔ x = a + kn, where k is an integer
```

Example: x ≡ 2 (mod 5) means x could be …, –8, –3, 2, 7, 12, 17, 22, … The condition describes infinitely many possible integers, not just x = 2.

3. What problem does CRT solve?

Suppose an unknown integer x satisfies all three conditions:

```
x ≡ 2 (mod 3)
x ≡ 3 (mod 5)
x ≡ 2 (mod 7)
```

The first condition allows 2, 5, 8, 11, 14, 17, 20, 23, …; the second allows 3, 8, 13, 18, 23, …; the third allows 2, 9, 16, 23, … The first common value is 23.

```
23 mod 3 = 2; 23 mod 5 = 3; 23 mod 7 = 2
```

CRT provides a systematic way to find the common value without guessing.

4. The theorem and its conditions

Chinese Remainder Theorem: If the moduli n■, n■, …, n■ are pairwise coprime, then the system x ≡ a■ (mod n■) has exactly one solution modulo N = n■n■…n■.

Pairwise coprime means every pair of different moduli has greatest common divisor 1. For example, 5, 11, and 17 are pairwise coprime.

“Unique modulo N” means there is exactly one answer from 0 to N – 1. All other integer solutions differ by a multiple of N. The moduli do not have to be prime; pairwise coprimality is sufficient.

5. Worked example using substitution

Solve the system:

```
x ≡ 2 (mod 5), x ≡ 3 (mod 11), x ≡ 5 (mod 17)
```

Step 1 — Start with the largest modulus

From x ≡ 5 (mod 17), write:

```
x = 5 + 17k
```

This represents every integer that leaves remainder 5 when divided by 17.

Step 2 — Apply the modulus 11

Substitute x = 5 + 17k into x ≡ 3 (mod 11):

```
5 + 17k ≡ 3 (mod 11)
```

Since 17 ≡ 6 (mod 11), simplify and subtract 5:

```
5 + 6k ≡ 3 (mod 11) ⇒ 6k ≡ 9 (mod 11)
```

The inverse of 6 modulo 11 is 2, because 6 × 2 = 12 ≡ 1 (mod 11). Multiply both sides by 2:

```
k ≡ 18 ≡ 7 (mod 11)
```

Thus k = 7 + 11t for an integer t. Substitute this into x = 5 + 17k:

```
x = 5 + 17(7 + 11t) = 124 + 187t
```

Step 3 — Apply the modulus 5

Use x ≡ 2 (mod 5):

```
124 + 187t ≡ 2 (mod 5)
```

Since 124 ≡ 4 and 187 ≡ 2 modulo 5:

```
4 + 2t ≡ 2 (mod 5) ⇒ 2t ≡ 3 (mod 5)
```

The inverse of 2 modulo 5 is 3, because 2 × 3 = 6 ≡ 1 (mod 5). Therefore:

```
t ≡ 9 ≡ 4 (mod 5)
```

Choose the smallest non-negative value t = 4:

```
x = 124 + 187 × 4 = 872
```

The product of the moduli is N = 5 × 11 × 17 = 935. Since 0 ≤ 872 < 935, the required smallest non-negative solution is:

```
x ≡ 872 (mod 935)
```

Verification: 872 mod 5 = 2; 872 mod 11 = 3; 872 mod 17 = 5. All conditions are satisfied.

6. The standard CRT formula

For pairwise-coprime moduli, calculate N = product of all moduli. For each modulus n■, define N■ = N / n■. Then find M■, the modular inverse of N■ modulo n■, so that N■M■ ≡ 1 (mod n■).

```
x ≡ Σ(a■ × N■ × M■) (mod N)
```

For the worked example, N = 935:

n■	a■	N■ = 935/n■	M■ = inverse of N■ modulo n■
5	2	187	3, because 187 mod 5 = 2 and 2 × 3 ≡ 1 mod 5
11	3	85	7, because 85 mod 11 = 8 and 8 × 7 ≡ 1 mod 11
17	5	55	13, because 55 mod 17 = 4 and 4 × 13 ≡ 1 mod 17

```
x ≡ 2×187×3 + 3×85×7 + 5×55×13 (mod 935)
x ≡ 1122 + 1785 + 3575 = 6482 (mod 935)
6482 mod 935 = 872
```

Both methods produce the same answer. The substitution method is often easier for beginners; the formula is convenient for larger systems.

7. Modular inverses

A modular inverse of a modulo n is a number b such that ab ≡ 1 (mod n). It exists exactly when gcd(a, n) = 1.

Examples: inverse of 2 modulo 5 is 3; inverse of 6 modulo 11 is 2. Always verify an inverse by multiplying and reducing modulo n. Do not treat modular division like ordinary division unless the inverse exists.

8. Python implementation

This implementation uses Python's built-in gcd and modular inverse. It assumes pairwise-coprime moduli. Python 3.8 or newer supports `pow(a, -1, n)` for modular inverses.

```
from math import gcd

def crt(remainders, moduli):
    if len(remainders) != len(moduli) or not moduli:
        raise ValueError("Inputs must have equal, non-zero lengths")

    for i, n in enumerate(moduli):
        if n <= 1:
            raise ValueError("Each modulus must be > 1")
        for m in moduli[i + 1:]:
            if gcd(n, m) != 1:
                raise ValueError("Moduli must be pairwise coprime")

    N = 1
    for n in moduli:
        N *= n

    x = 0
    for a, n in zip(remainders, moduli):
        Ni = N // n
        Mi = pow(Ni, -1, n)
        x += a * Ni * Mi

    return x % N

answer = crt([2, 3, 5], [5, 11, 17])
print(answer)  # 872

assert all(answer % n == a for a, n in zip(
    [2, 3, 5], [5, 11, 17]
))
```

For your MathForge project, replace gcd and modular-inverse calls with the verified MathForge APIs. Check the installed function signatures first; do not assume an undocumented CRT function exists.

9. Why CRT matters in cryptography CTFs

RSA optimization: RSA implementations can compute a private operation modulo p and q separately, then recombine the results with CRT.

RSA broadcast attack: When the same unpadded message is encrypted under a small public exponent and pairwise-coprime moduli, CRT can combine ciphertexts. Recovering the message still depends on the required size and padding assumptions.

Secret reconstruction: If a challenge gives x modulo several coprime numbers, CRT recovers x modulo their product. Extra bounds may be needed to identify the original integer.

Important: CRT is a mathematical tool, not an attack by itself. It does not automatically factor RSA moduli or break secure encryption.

10. What if the moduli are not coprime?

The standard version above assumes pairwise-coprime moduli. For two congruences x ≡ a (mod m) and x ≡ b (mod n), a solution exists exactly when a ≡ b (mod gcd(m,n)). If compatible, the solution is unique modulo lcm(m,n).

```
Example: x ≡ 1 (mod 4), x ≡ 3 (mod 6)
```

Here gcd(4,6) = 2 and 1 ≡ 3 (mod 2), so a solution exists; x = 9 is one solution. By contrast, x ≡ 1 (mod 4), x ≡ 2 (mod 6) has no solution because the remainders disagree modulo 2.

11. Common mistakes

• Thinking x ≡ a (mod n) means x = a; it describes infinitely many integers.

• Using a modular inverse without checking that it exists.

• Calculating an inverse incorrectly; verify N■M■ mod n■ = 1.

• Forgetting to reduce the final result modulo N.

• Assuming every non-coprime system has a solution.

• Trusting code output without checking every original congruence.

12. Practice problems

1. x ≡ 1 (mod 3), x ≡ 2 (mod 5). Find the smallest non-negative solution.

2. x ≡ 2 (mod 3), x ≡ 3 (mod 5), x ≡ 1 (mod 7). Find x modulo 105.

3. x ≡ 1 (mod 4), x ≡ 2 (mod 6). Does a solution exist? Explain using gcd.

Answers: 1) 7 modulo 15. 2) 8 modulo 105. 3) No; gcd(4,6)=2, but 1 and 2 are not congruent modulo 2.

13. Quick revision sheet

• Meaning: x ≡ a (mod n) means n divides (x – a).

• Pairwise coprime: gcd(n■,n■)=1 for every i ≠ j.

• Combined modulus: N = n■n■…n■.

• Partial product: N■ = N/n■.

• Inverse: M■ = N■■¹ mod n■.

• CRT formula: x ≡ Σ(a■N■M■) mod N.

• Always verify x mod n■ = a■ mod n■ for every equation.

Memory trick: Remainders → Product → Partial products → Inverses → Weighted sum → Reduce → Verify.